

Data Science

Workshop 2: Introduction to NumPy

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Activities

- ▶ Moodle Submission
 - ▶ Single document with Jupyter screenshots
 - ▶ Computation on NumPy Arrays
 - ▶ Indexing & sorting in NumPy
 - ▶ Probability distribution in NumPy
 - ▶ Structured data using NumPy

Computation On NumPy Arrays (Part 1 of 5)

- ▶ `#Start jupyter notebook`
- ▶ `import numpy as np`
- ▶ `#Create a 2D array`
- ▶ `#Print the array`

Computation on NumPy Arrays (Part 2 of 5)

- ▶ `import numpy as np`
- ▶ `#Create 2 same size numeric arrays`
- ▶ `#Add the 2 arrays`
- ▶ `#Print the result`
- ▶ `#Subtract one array from the other`
- ▶ `#Print the result`

Computation on NumPy Arrays (Part 3 of 5)

- ▶ `import numpy as np`
- ▶ `#Create a numeric array`
- ▶ `#Use the prod function`
- ▶ `#Print the result`
- ▶ `#Use the diff function`
- ▶ `#Print the array`

Computation on NumPy Arrays (Part 4 of 5)

- ▶ `import numpy as np`
- ▶ `#Try the log function`
- ▶ `#Print the result`
- ▶ `#Try the log2 function`
- ▶ `#Print the result`
- ▶ `#Try the sine function`
- ▶ `#Print the result`

Computation on NumPy Arrays (Part 5 of 5)

- ▶ `from numpy import random`
- ▶ `import numpy as np`
- ▶ `#Create a random number from 0 to 1000`
- ▶ `#Print the result`
- ▶ `arr = np.array(5, 2, 1, 3, 0])`
- ▶ `#Shuffle the array`
- ▶ `#Print the result`
- ▶ `#Shuffle the array again`
- ▶ `# Print the result`

Indexing & Sorting NumPy

- ▶ `from numpy import random`
- ▶ `import numpy as np`
- ▶ `arr = np.array(5, 2, 1, 3, 0])`
- ▶ `#Sort the array`
- ▶ `# Print the result`
- ▶ `#Slice the last two elements`
- ▶ `#Print the result`

Probability Distribution in NumPy

- ▶ `#Create a binormal distribution plot using with  $p = 0.3$`
- ▶ `#Print the graphic result`

Moodle Submission

Structured Data using NumPy

- ▶ #Follow the commands on <https://jakevdp.github.io/PythonDataScienceHandbook/02.09-structured-data-numpy.html> from Step [1] to Step [13] and print the result.